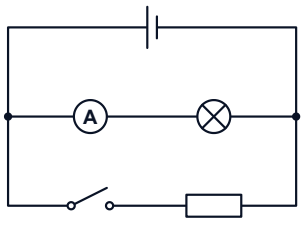
DO NOW, AO1 REVISION

Series and parallel circuits

Electricity
Paper 1, Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the two ways components can be joined in a circuit.**	In series and in parallel	☐
2 ★	**Describe the current through components in series.**	It is the same through each component.	☐
3	**Describe the potential difference across components in series.**	The total potential difference is shared between them.	☐
4	**Describe the total resistance of components connected in series.**	It is the sum of the individual resistances.	☐
5	**Describe the current through components in parallel.**	The total current is the sum of the currents in the branches.	☐
6	**The lamp in this circuit is lit. Describe what happens to the ammeter reading when the switch is closed.** 	It does not change. The lamp is on a separate branch, so the potential difference across it stays the same.	☐
7 ★	**Describe the potential difference across components in parallel.**	It is the same across each component.	☐
8 ★	**Describe the total resistance of two resistors connected in parallel.**	It is less than the resistance of the smaller resistor.	☐
9	**Explain why adding resistors in series increases the total resistance.**	The potential difference is shared between more resistors, so the current is smaller.	☐
10 ★	**Explain why adding resistors in parallel decreases the total resistance.**	Each new branch gives the charge another path, so the total current increases.	☐

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	QUESTION	ANSWER	✓
11	**Lamps are connected in series and one breaks. State what happens to the others.**	They all go out, because the circuit is broken.	☐
12	**Give two advantages of connecting lamps in parallel rather than in series.**	Each lamp can be switched separately, and if one breaks the others stay lit.	☐

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